Do Corporate Disclosures Improve Intraday Forecasts? Evidence from a Point-in-Time China A-Share Panel

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Abstract

We evaluate whether public corporate disclosures improve short-horizon stock-return forecasts after a strong intraday-price benchmark has been fixed. At 11:30 China Standard Time, the task is to forecast whether each stock's 15:00 close will exceed its 11:30 close. We construct a point-in-time CSI 300 panel with 479,540 stock-day observations, 493 historical constituents, and 1,601 trading dates from 2020 to 2026. The price input contains 24 five-minute morning bars. We also collect 435,374 CNInfo announcements and impose a conservative availability rule when a record identifies only its publication date.

A five-seed temporal convolutional network (TCN) ensemble using prices alone obtains a primary-holdout AUC of 0.5513, compared with 0.5275 for the strongest prespecified summary-feature baseline. Adding structured disclosure features raises the point estimate to 0.5538, but the paired difference is not significant. In a prespecified five-fold expanding-window robustness analysis covering 209,670 out-of-fold observations and 700 non-overlapping test dates, price TCN AUC is 0.5482 versus 0.5223 for the baseline; the difference is 0.0259 with a trading-date-clustered 95% bootstrap interval of $[0.0168, 0.0350]$. Disclosure fusion instead reaches 0.5460, a difference of -0.0022 versus prices with interval $[-0.0056, 0.0011]$ ($p = 0.210$), and no predefined market regime shows a reliable disclosure increment. Filing-body coverage reaches 98.73%, but frozen text representations also add no reliable signal. Top-decile rolling portfolios earn 2.47 basis points per day gross for the price model and become negative after a five-basis-point round-trip cost. The evidence supports a reproducible but economically weak price-sequence signal and no robust incremental value from the tested disclosure representations.

Keywords: intraday forecasting; corporate disclosures; China A-share market; temporal convolutional network; point-in-time evaluation; economic value

1 Introduction

Forecasts built from corporate disclosures are attractive because filings contain information about earnings, contracts, financing, governance, litigation, and other firm-specific events. Yet their incremental value over contemporaneous prices is difficult to establish. Public information may already be reflected in prices, publication timestamps may be imprecise, and observations for different stocks on the same trading day share common shocks. Small reported gains can also depend on a favorable neural-network seed, an ex-post stock universe, or test-set-guided model selection.

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We study these issues in a short-horizon China A-share forecasting task. At the end of the morning trading session, 11:30 China Standard Time, we forecast the sign of the return from the observed 11:30 close to the 15:00 close. The information set contains the current morning's five-minute price path and corporate announcements verifiably available after the previous trading close and no later than the forecast cutoff. Historical CSI 300 membership is applied using official effective dates rather than backfilling the current constituent list.

The study asks three questions. First, does a temporal model of the morning price path improve forecasts over fixed summary-feature baselines? Second, after that price model has been selected, do structured announcement variables, titles, or filing bodies add out-of-sample ranking information? Third, do any predictive differences translate into economically meaningful returns when execution begins at the first tradable afternoon-session price and transaction costs are imposed?

The evidence separates these questions. Morning price sequences contain a small but reproducible signal: a five-seed TCN probability ensemble reaches 0.5513 on the primary holdout and significantly exceeds the strongest summary-feature baseline. Across five expanding-window folds, the corresponding out-of-fold AUC is 0.5482 versus 0.5223 for HGB. Structured disclosures increase the primary-holdout AUC point estimate by 0.0025, but the clustered confidence interval includes zero; over the five rolling folds, disclosure fusion is instead 0.0022 below prices. Neither the announcement-only subset nor any predefined market regime supplies a reliable positive increment. Title models are weak, and expanding body-text coverage to 98.73% does not improve the conclusion. Simple top-score portfolios have positive gross point estimates but become negative after modest costs.

The contribution is therefore primarily an evaluation result rather than a claim of a new forecasting architecture. We provide (i) a six-year point-in-time panel that avoids current-constituent backfill; (ii) a conservative treatment of disclosure-time precision; (iii) an incremental design that freezes a price benchmark before testing disclosure representations; (iv) multi-seed, expanding-window, market-regime, and trading-date-clustered inference; and (v) a separation of predictive ranking, statistical uncertainty, and executable economic value. The high-coverage null text result is informative because missing filing bodies no longer provide a plausible explanation for the absence of incremental performance.

2 Related Literature

2.1 Financial text and disclosure-based prediction

Financial text prediction has progressed from early evidence that media pessimism and firm-specific language relate to returns and fundamentals (Tetlock, 2007; Tetlock et al., 2008) to context-sensitive news features (Hagenau et al., 2013), structured event representations (Ding et al., 2014, 2015), word embeddings (Peng and Jiang, 2016), joint text–price systems (Xu and Cohen, 2018), and finance-domain language models (Araci, 2019; Yang et al., 2020). Domain-specific wording matters because general dictionaries can misclassify common financial terms (Loughran and McDonald, 2011); the broader accounting and finance literature likewise emphasizes implementation choices and limited precision in textual measures (Loughran and McDonald, 2016). Regulatory filings are especially relevant to the present setting because their release can be tied to a formal issuer event. Prior work reports short-horizon information in full filing text (Lee et al., 2014), while evidence from China relates news tone to equity returns (Ge and Zhang, 2022).

Our focus differs from comparisons that search for the best standalone text model. We estimate the increment over an identical-sample price benchmark, distinguish title, structured-event, and body-text signals, and keep observations without disclosures in the primary panel. This makes the

estimand operational: whether adding the available disclosure channel changes forecasts for the stock-day universe faced by the price model.

2.2 Forecast evaluation and financial model selection

Small financial forecast improvements require unusually careful evaluation. Rapid incorporation of public information makes a weak or null disclosure increment plausible under market efficiency (Fama, 1970). Repeated strategy or architecture searches create backtest-overfitting risk (Bailey et al., 2017). Classical forecast-comparison work formalizes equal predictive accuracy and conditional predictive ability (Diebold and Mariano, 1995; Giacomini and White, 2006). Chronological training and validation are therefore essential for dependent time-series data (Bergmeir et al., 2018), while rolling origins and multiple test periods reduce sensitivity to a particular split or market phase (Tashman, 2000). Cross-sectional stock observations within a date are also dependent, motivating date-clustered rather than observation-level uncertainty estimates (Cameron et al., 2008).

For the sequence benchmark, we compare recurrent and convolutional architectures rather than assume that greater complexity ensures stronger forecasts. TCNs provide a robust general sequence baseline (Bai et al., 2018); InceptionTime is a useful time-series classification reference (Fawaz et al., 2020); and recent evidence cautions that complex attention architectures do not automatically dominate simpler forecasting models (Zeng et al., 2023). In empirical asset pricing, flexible machine-learning models can capture nonlinear interactions, but their value is established through comparative out-of-sample performance rather than architectural novelty alone (Gu et al., 2020). The A-share setting has distinctive institutions and information environments (Carpenter and Whitelaw, 2017) and further requires attention to price limits, the $T+1$ trading cycle, and market-specific execution frictions (Peng et al., 2024).

3 Data and Forecasting Task

3.1 Point-in-time price panel

The raw BaoStock collection contains 36,457,410 five-minute rows for 493 securities that appeared in the CSI 300 during the study period. Applying official constituent effective dates leaves 23,027,980 rows. Requiring a complete 24-bar morning sequence and a valid afternoon target produces 479,540 stock-day observations across 1,601 trading dates from January 2, 2020 through August 11, 2026.

For stock i on trading day t, the target is

$$y_{i,t} = \mathbb{1}\left(P_{i,t}^{15:00} / P_{i,t}^{11:30} - 1 > 0\right). \tag{1}$$

The model observes only the 24 five-minute bars ending at 11:30. Sequence variables are cumulative return from the morning open, bar-to-bar return, log volume, log turnover amount, and the high–low range. Tabular baselines use morning return, realized volatility, morning range, log amount, last-30-minute return, and volume-weighted average price.

Table 1: Frozen chronological split.

Split	Final date	Trading dates	Observations
Training	2024-08-14	1,120	335,552
Validation	2025-08-12	240	71,844
Test	2026-08-11	241	72,144

3.2 Corporate disclosures and information availability

We collect 435,374 unique CNInfo announcements using 3,451 stock-year requests. All requests completed without an unresolved failure. For every stock-day, an announcement is eligible only when it becomes available strictly after the preceding observed trading close at 15:00 and no later than 11:30 on the prediction day.

Timestamp precision is a central design issue. CNInfo supplies an exact within-day timestamp for 31,817 records, whereas 403,557 historical records are encoded only as a calendar date at midnight. Interpreting those values as verified midnight releases would introduce false availability. The primary policy therefore assigns date-only records an availability time of 16:00 on the listed date, after market close; the announcement can first enter the next eligible trading-day window. A prespecified sensitivity excludes all date-only records.

Under the primary policy, 271,062 announcements map to 86,898 stock-days, or 18.1% of the panel. The test set contains 12,830 observations with at least one eligible announcement. Exact-time-only coverage is substantially smaller: 8,083 observations overall and 1,380 in test.

For the full-text extension, a title-only rule fixed before body-model evaluation selects 39,667 primary high-signal announcements across 28,651 eligible stock-days. Repetitive supporting valuation, adviser, verification, opinion, and incentive-list attachments are excluded. Audited PDF acquisition yields at least 200 normalized characters for 39,164 announcements (98.73%); 500 short or image-only filings and three failures remain in the denominator and are represented with explicit missing-body features.

4 Experimental Design

4.1 Price benchmarks and architecture selection

Logistic regression and histogram gradient boosting (HGB) operate on the six morning summary variables. Neural candidates are a one-layer LSTM, a two-layer one-dimensional CNN with average pooling, and a TCN with three residual temporal blocks and dilations 1, 2, and 4. All use hidden width 64, dropout 0.2, AdamW with learning rate 10^{-3} and weight decay 10^{-4}, batch size 1,024, and early stopping on validation AUC with patience five. Positive-class weights are estimated from training data only.

Seed 42 is used for architecture selection using validation AUC. The selected architecture is then rerun with seeds 7, 21, 42, 84, and 2026. The prespecified ensemble averages the five predicted probabilities. No architecture is selected on test AUC.

4.2 Disclosure representations and fusion

Structured features include announcement and event-category counts, title-length statistics, polarity and uncertainty keywords, earnings guidance, buybacks, shareholder changes, contracts, litigation, pledges, regulatory inquiries, suspensions, dividends, restructuring, directional cues, and the presence of percentages, amounts, share counts, and durations. The sequence-fusion model concatenates the TCN representation with a two-layer disclosure branch before the final prediction head.

Title and body text are evaluated separately. Character TF–IDF uses 2–4 character n-grams and a logistic-loss linear classifier. The semantic representation uses frozen BGE-small-zh-v1.5 embeddings (Xiao et al., 2024) followed by a training-only logistic classifier. Body input combines the title and up to three extractively selected sentences per announcement. A frozen Chinese FinBERT-tone model supplies positive, neutral, and negative probability diagnostics. Global weighted, announcement-gated, and meta-logistic late fusion are selected by validation AUC only.

Samples without an eligible announcement receive a neutral text score, and unusable bodies are never silently removed.

4.3 Inference and robustness

ROC AUC is the primary predictive metric; balanced accuracy, accuracy, and F1 at threshold 0.5 are secondary. Model comparisons use paired predictions on identical observations. Uncertainty is estimated with 5,000 paired bootstrap draws clustered by trading date. The cluster design preserves contemporaneous cross-stock dependence that an observation-level bootstrap would ignore.

The label analysis is repeated after excluding exactly zero returns and after excluding observations within 5- and 10-basis-point neutral bands. For disclosure models, results are reported on both the full panel and the subset containing at least one eligible announcement. Exact-timestamp-only availability and complete-body text are additional sensitivities.

After the primary holdout had been analyzed, but before inspecting any rolling result, we froze an expanding-window robustness protocol. The initial training window contains 761 trading dates. Each of five folds uses the next 140 dates for validation and the following 140 dates for testing; the origin then advances by 140 dates, so test windows do not overlap. The five neural seeds and all optimization settings remain unchanged in every fold. This produces 209,670 out-of-fold test observations across 700 dates from September 18, 2023 through August 11, 2026. Because this protocol was motivated by the primary analysis, it is treated as temporal robustness rather than as a new independent confirmation sample.

Market regimes are also fixed before rolling results are read and use only information available by 11:30. Market direction is the sign of the cross-sectional mean morning return. High market volatility and high cross-sectional dispersion exceed the respective 70th percentiles estimated from each fold's training dates; high trading activity exceeds the training-date median. Disclosure availability separates stock-days with and without an eligible announcement. Each regime comparison retains the paired five-seed price and disclosure ensembles and resamples complete trading dates.

4.4 Executable economic evaluation

The statistical target begins at the already observed 11:30 close, so it cannot be used as a post-signal execution price. Economic evaluation instead enters at the 13:00 afternoon-session open, defined as the open of the five-minute bar ending at 13:05, and exits at 15:00. Entry requires a positive price and volume and a complete 24-bar afternoon sequence. Each day, the portfolio equally weights stocks in the top 10% of model scores; top 5% and top 20% are breadth sensitivities. We report gross returns and 5-, 10-, 20-, and 30-basis-point round-trip cost scenarios. Daily portfolio returns, rather than stock observations, are the bootstrap units.

4.5 Use of generative AI tools

During the preparation of this manuscript, the authors used OpenAI ChatGPT and Codex for language editing, code review, analytical workflow support, and manuscript organization. The authors reviewed and verified all outputs and take full responsibility for the content of the manuscript.

5 Results

5.1 Price-sequence forecasts

At seed 42, validation AUC is 0.5391 for the LSTM, 0.5371 for the CNN, and 0.5398 for the TCN. The TCN is therefore selected without consulting test performance. Its five-seed test AUC is 0.5478 ± 0.0018, and probability averaging raises test AUC to 0.5513. The HGB summary baseline reaches 0.5275 and logistic regression reaches 0.5116.

The paired test AUC difference between the TCN ensemble and HGB is 0.0238. Its trading-date-clustered 95% interval is $[0.0110, 0.0370]$ with $p = 0.0012$. Thus, the morning path contributes ranking information beyond the prespecified summaries, although the absolute forecast performance remains modest.

Table 2: Primary frozen-split predictive results. Neural means use five prespecified seeds; ensemble rows average their probabilities.

Model	Validation AUC	Test AUC	Test balanced accuracy
Logistic price summary	0.5051	0.5116	0.5088
HGB price summary	0.5139	0.5275	0.5018
HGB price + disclosure	0.5133	0.5258	0.5021
TCN price, five-seed mean	0.5399	0.5478	0.5306
TCN price ensemble	0.5426	0.5513	0.5355
TCN + disclosure, five-seed mean	0.5421	0.5507	0.5338
TCN + disclosure ensemble	**0.5443**	**0.5538**	0.5355

5.2 Incremental value of structured disclosures

Disclosure features do not improve the tabular HGB benchmark: test AUC declines from 0.5275 to 0.5258. Exact-timestamp-only disclosure HGB reaches 0.5260, also below price-only HGB.

The neural disclosure-fusion ensemble has a higher full-sample point estimate than the price ensemble, 0.5538 versus 0.5513. However, the paired gain of 0.0025 has a trading-date-clustered interval of $[-0.0002, 0.0051]$ and $p = 0.070$. Among the 12,830 test observations with at least one eligible announcement, price AUC is 0.5535 and fusion AUC is 0.5527. The difference is -0.0008, with interval $[-0.0067, 0.0046]$ and $p = 0.784$. The defensible conclusion is a borderline full-panel point estimate, not an established disclosure effect.

Table 3: Paired ensemble comparison using 5,000 trading-date-clustered bootstrap draws.

Subset	n	Price AUC	Fusion AUC	Difference [95% CI]
Full test	72,144	0.5513	0.5538	0.0025 [−0.0002, 0.0051]
Has announcement	12,830	0.5535	0.5527	−0.0008 [−0.0067, 0.0046]

5.3 Expanding-window and market-regime robustness

Table 4 reports the five non-overlapping rolling test windows. Price TCN exceeds HGB in every fold. Four fold-level clustered intervals exclude zero; fold 3 is borderline ($p = 0.062$). Pooling only the out-of-fold test predictions gives HGB AUC 0.5223 and price TCN AUC 0.5482, a difference of

0.0259 with a 95% interval of $[0.0168, 0.0350]$. No bootstrap draw produces a non-positive pooled difference; given 5,000 draws, we report $p < 0.0004$ rather than zero.

Disclosure fusion does not share this robustness. It is below price TCN in the first three folds and nearly equal in the last two. The pooled difference is -0.0022, with interval $[-0.0056, 0.0011]$ and $p = 0.210$. None of the five fold-level disclosure differences is significant.

Table 4: Five-fold expanding-window out-of-fold results. Each test fold contains 140 non-overlapping trading dates.

Fold	n	HGB AUC	Price TCN AUC	Fusion AUC	Fusion − price
1	41,995	0.5360	0.5598	0.5566	−0.0032
2	41,923	0.5062	0.5351	0.5273	−0.0078
3	41,915	0.5279	0.5453	0.5418	−0.0035
4	41,875	0.5291	0.5510	0.5528	0.0019
5	41,962	0.5241	0.5513	0.5514	0.0001
Pooled OOF	209,670	0.5223	0.5482	0.5460	−0.0022

Table 5 shows the predefined market-state comparisons. All ten clustered intervals include zero. In particular, the disclosure model does not improve when an eligible announcement is present, in high-volatility or high-dispersion mornings, or when market activity is high. The regime analysis therefore provides no evidence that a favorable aggregate effect is hidden in an economically obvious state.

Table 5: Disclosure-fusion increment by predefined rolling-test regime. Cutoffs are estimated from each fold's training dates.

Regime	n	Price AUC	Fusion AUC	Difference [95% CI]
Market down	98,555	0.5536	0.5495	−0.0041 [−0.0096, 0.0011]
Market up	111,115	0.5422	0.5413	−0.0009 [−0.0050, 0.0031]
Low market volatility	176,106	0.5462	0.5454	−0.0008 [−0.0038, 0.0021]
High market volatility	33,564	0.5596	0.5493	−0.0102 [−0.0246, 0.0027]
Low cross-sectional dispersion	168,917	0.5474	0.5454	−0.0020 [−0.0054, 0.0014]
High cross-sectional dispersion	40,753	0.5555	0.5513	−0.0042 [−0.0156, 0.0059]
Low trading activity	63,823	0.5461	0.5436	−0.0025 [−0.0078, 0.0027]
High trading activity	145,847	0.5486	0.5461	−0.0024 [−0.0068, 0.0016]
No eligible announcement	172,796	0.5467	0.5449	−0.0018 [−0.0053, 0.0015]
Eligible announcement	36,874	0.5548	0.5516	−0.0032 [−0.0082, 0.0015]

5.4 Title, body-text, and feature diagnostics

Character TF–IDF titles obtain test AUC 0.5062 on announcement observations, while frozen BGE title embeddings reach 0.5129. Validation-selected global BGE late fusion changes full-test price AUC by only 0.000093; the clustered interval is $[-0.000353, 0.000525]$ with $p = 0.653$. Announcement-gated validation selects a text weight of zero.

High-coverage filing bodies do not reverse this result. On 4,055 test observations with selected disclosures, standalone TF–IDF, frozen BGE, and FinBERT-tone AUCs are 0.4982, 0.5054, and 0.5034. The validation-selected TF–IDF meta fusion reaches 0.551111 on the full test set, versus 0.551290 for prices. The difference is -0.000179 with clustered interval $[-0.000860, 0.000537]$ and $p = 0.628$. On disclosure observations, the difference is -0.007492 with interval $[-0.016268, 0.001206]$ and $p = 0.091$.

Table 6: Frozen filing-body results. Standalone AUC is evaluated on test observations with selected disclosures; fusion differences are relative to the price TCN ensemble.

Body representation	Standalone AUC	Full-test difference	p
Character TF–IDF	0.4982	−0.000179	0.628
Frozen BGE	0.5054	−0.000012	0.162
FinBERT tone	0.5034	0.000000	–

Separate HGB diagnostics add disclosure coverage and counts, polarity keywords, event types, and direction or numeric cues. None exceeds price-only HGB on the full test set. Eight event-category subgroup tests, including earnings guidance, buybacks, contracts, dividends, pledges, and restructuring, yield no Benjamini–Hochberg-adjusted significant category. These results do not justify selecting a favorable event group after observing test outcomes.

5.5 Neutral-return and economic-value robustness

Retraining the seed-42 TCN after excluding exact-zero, 5-basis-point-neutral, and 10-basis-point-neutral observations yields test AUCs of 0.5454, 0.5436, and 0.5470. The signal therefore persists across label conventions, but its magnitude varies enough that the original binary definition should not be presented as uniquely correct.

Using the executable 13:00 entry changes the return direction for 2.77% of observations relative to the 11:30-close target. Over 241 primary-test dates, the price and structured-disclosure top-decile portfolios earn 4.26 and 4.60 basis points per day gross. Their paired difference is only 0.33 basis points per day, with a date-bootstrap interval of $[-1.70, 2.29]$. At five basis points round trip, mean daily returns become −0.74 and −0.40 basis points.

The longer rolling out-of-fold evaluation is weaker economically. Across 700 test dates, top-decile price and disclosure portfolios earn 2.47 and 1.54 basis points per day gross; both gross confidence intervals include zero. The disclosure-minus-price gross difference is −0.93 basis points per day with interval $[-2.24, 0.41]$. After a five-basis-point round-trip cost, the means are −2.53 and −3.46 basis points per day. Top-5% and top-20% breadth checks reach the same qualitative conclusion, and losses deepen at 10–30 basis points. Full-text portfolio differences also have intervals including zero.

Table 7: Executable top-decile long-only portfolios. Entry is 13:00 and exit is 15:00; annualized returns are arithmetic means multiplied by 252.

Evaluation	Model	Gross bps/day	Gross annualized	Net bps/day at 5 bps
Primary holdout	Price TCN	4.26	10.74%	−0.74
Primary holdout	Disclosure fusion	4.60	11.66%	−0.40
Rolling OOF	Price TCN	2.47	6.23%	−2.53
Rolling OOF	Disclosure fusion	1.54	3.87%	−3.46

6 Discussion

The main empirical pattern is asymmetric. Intraday price paths add reproducible ranking information beyond summary variables, whereas corporate disclosures do not add a reliable increment beyond those paths. The five expanding-window folds strengthen the first conclusion and weaken any

interpretation of the primary disclosure point estimate: the price advantage survives different validation and test periods, while the disclosure increment is negative in pooled rolling predictions and absent in every predefined regime. Several mechanisms could produce the disclosure result. Public announcements may be incorporated before the forecast cutoff; generic semantic embeddings may not represent filing-specific quantities and direction well; and date-only publication records force a conservative delay that trades leakage risk for reduced immediacy. The result therefore concerns the tested point-in-time information policy and frozen representations, not the proposition that disclosures never matter.

The announcement-only result is particularly important. A small positive full-panel gain accompanied by a negative announcement-subset estimate is not strong evidence that announcement content drives the difference. Sparse structured indicators may change regularization or calibration across the whole panel. Because the clustered interval includes zero, we avoid assigning that point estimate an economic mechanism.

The full-text experiment also changes the interpretation of the null result. With only partial body coverage, weak performance could be attributed to missing filings. At 98.73% usable-body coverage, that explanation is no longer persuasive for the frozen event set. The remaining possibilities concern representation, precise timing, the forecast horizon, or genuinely limited incremental information. A future domain-adapted encoder must be trained using training-period text only and evaluated under a newly frozen validation and test protocol; tuning it against the present test period would convert the current test set into development data.

The study also illustrates why statistical and economic claims must remain separate. AUC evaluates cross-sectional ranking over stock observations, whereas an implementable portfolio begins after the forecast is produced and depends on turnover, price limits, queues, taxes, and market impact. Positive gross point estimates are insufficient when paired differences are uncertain and modest transaction costs remove average returns.

7 Limitations and Further Research

The disclosure source covers formal corporate announcements but not every market-moving news channel. Most historical CNInfo records provide only a date, and the conservative 16:00 policy may delay documents that were actually public earlier. Exact-timestamp sensitivity has much lower coverage and therefore limited power. Filing extraction is capped at ten pages and image-only documents are represented as missing. The frozen text encoders are not fine-tuned for the task.

The economic simulation is deliberately simple. It does not fully reproduce price-limit queues, capacity, stamp duty, stock-specific impact, intrabar slippage, or rejected orders. It should be interpreted as an executable-price diagnostic rather than a deployable strategy backtest.

The five expanding-window tests reduce dependence on one terminal holdout, but they are not an external replication. They were motivated after the primary holdout had been inspected, reuse overlapping training histories, and cover the same market and data provider. Their protocol was frozen before rolling outcomes were read, which limits result-driven specification changes but does not restore the status of a wholly independent confirmation sample. Evidence from a later period, another Chinese equity universe, or another market would provide a stronger external test.

8 Conclusion

On a six-year point-in-time China A-share panel, morning five-minute price sequences provide a small but statistically distinguishable forecast improvement over summary-feature baselines. The

price-model advantage persists across five non-overlapping expanding-window tests, with pooled out-of-fold AUC 0.5482 versus 0.5223 for HGB. Structured corporate disclosures raise the primary-holdout neural AUC by 0.0025, but the interval crosses zero; in rolling evaluation, fusion is 0.0022 below prices and no predefined market regime improves reliably. Neither titles nor 98.73%-coverage filing bodies add a reliable increment under the frozen representations. Gross portfolio returns are uncertain and become negative under a five-basis-point cost scenario.

The evidence supports a cautious forecasting conclusion: evaluation choices materially affect the apparent value of multimodal financial signals. Conservative information timing, point-in-time universes, multiple seeds, dependence-aware inference, and executable prices should be treated as part of the forecasting method rather than as optional robustness checks.

Data Availability Statement

The derived feature tables, frozen split definitions, prediction files, and code required to reproduce the reported tables will be made available in a public repository upon publication, subject to the redistribution terms of BaoStock and CNInfo. Raw third-party market data and filing PDFs are not redistributed where their source terms do not permit it. A review archive with checksums, extraction audit records, and instructions for rebuilding the data will be provided.

Code Availability

Training, evaluation, clustered-bootstrap, and economic-value scripts are archived at https://github.com/john092985/a-share-intraday-disclosure-forecasting. The repository is private during manuscript finalization and will be made public with the submitted or accepted version after a final review of third-party provider terms.

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Conflict of Interest

The authors declare no conflicts of interest.

Author Contributions

Jingxuan Lyu: Conceptualization, methodology, software, formal analysis, data curation, visualization, and writing—original draft. Xueshi Bai: Validation and writing—review and editing. Both authors reviewed and approved the final manuscript.

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